

① Parameter Estimation

$X_1, X_2, \dots, X_n \sim \text{i.i.d distribution depending on } \theta$
Goal: estimate unknown parameters.

① Frequentist approach: θ as an unknown const.

$\rightarrow$ Method of moments

$\rightarrow$ Max-Likelihood Estimation (MLE)

② Bayesian Approach: θ as a random variable

$\rightarrow$ has a distribution (called prior)

Bayesian Estimation $\rightarrow P(\theta|s) = \frac{P(s|\theta) \cdot P(\theta)}{P(s)}$

Posterior $\leftarrow$ Likelihood $\times$ Prior

Normalizing constant $\leftarrow P(s)$

Q. Let $X_1, \dots, X_n \sim \text{i.i.d Bernoulli}(p)$ distribution.
let p has the following PMF:

p	0.3	0.7
$P(p)$	0.8	0.2

Consider the samples 1, 1, 0, 1, 0
from Bernoulli (p)

$\Rightarrow$ Find the posterior Mode

$X_1, \dots, X_n \sim \text{Bernoulli}(p)$
Prior $P(p)$: $P(p=0.3) = 0.8$
 $P(p=0.7) = 0.2$

① Count data: No. of 1s: $w=3$

No. of 0s: $n-w=2$

② Likelihood: $P(S|p) = p^w (1-p)^{n-w}$

Case 1: $P=0.3$ $P(S|0.3) = (0.3)^3 (0.7)^2$

Case 2: $P=0.7$ $P(S|0.7) = (0.7)^3 (0.3)^2$

③ Posterior (Ignore denominator)

$P(p|S) \propto P(S|p) \cdot P(p)$

for $p=0.3$: $P(p=0.3|S) \propto (0.3)^3 (0.7)^2 (0.8)$

for $p=0.7$: $P(p=0.7|S) \propto (0.7)^3 (0.3)^2 (0.2)$

④ Compare (for mode)

Take ratio: $\frac{P(p=0.3|s)}{P(p=0.7|s)} = \frac{(0.3)^3 (0.7)^2 (0.8)}{(0.7)^3 (0.3)^2 (0.2)} = \frac{1.2}{0.7} \approx 1.71 > 1$

$\Rightarrow$ Posterior mode = 0.3

$\Rightarrow$ Find posterior mean?

$$E[p|s] = 0.3 \cdot P(0.3|s) + 0.7 \cdot P(0.7|s)$$

$$A = (0.3)^3 \cdot (0.7)^2 \cdot (0.8)$$

$$B = (0.7)^3 \cdot (0.3)^2 \cdot (0.2)$$

0.7 factor common term

$$A \propto 0.3 \cdot 0.8 = 0.24$$

$$B \propto 0.7 \cdot 0.2 = 0.14$$

Normalize: $P(0.3|s) = \frac{0.24}{0.24 + 0.14} = \frac{0.24}{0.38} \approx 0.6316$

~~Posterior mean~~ $P(0.7|s) = \frac{0.14}{0.38}$

Posterior Mean

$$E[p|s] = 0.3 \cdot \frac{0.24}{0.38} + 0.7 \cdot \frac{0.14}{0.38}$$

$$= \frac{0.17}{0.38} \approx 0.447$$

Priors

- Flat Prior $\rightarrow$
 - Uniform $\rightarrow$ every value is equally likely
 - No Prior knowledge.
- Conjugate Prior $\rightarrow$ Prior and Posterior belong to the same family.

Standard Conjugate Pairs (MEMORIZE)

<u>Likelihood</u>	<u>Prior</u>	<u>Posterior</u>
Bernoulli	Beta	Beta
Poisson	Gamma	Gamma
Normal(μ)	Normal	Normal
Exponential	Gamma	Gamma

Bernoulli + Beta ★ IMP.

Setup:

- $X_i \sim \text{Bernoulli}(p)$
- Prior: $p \sim \text{Beta}(\alpha, \beta)$

- Cases:
- $\alpha = \beta = 1 \rightarrow \text{Uniform}$
 - $\alpha = \beta \rightarrow \text{Symmetric}$
 - $\alpha = \beta = 0 \rightarrow \text{MLE}$

Posterior: $p|S \sim \text{Beta}(\alpha + w, \beta + n - w)$

Posterior-mean: $E[p|S] = \frac{\alpha + w}{\alpha + \beta + n}$

- $\alpha = \text{Prior successes}$
- $\beta = \text{Prior failures}$

ex: i.i.d. samples $\Rightarrow 1, 0, 0, 1, 1, 0, 1, 1, 0$
from Bernoulli(p). Let prior $p \sim \text{Beta}(\frac{1}{2}, \frac{1}{2})$.
find posterior distribution of p .

Sol. no. of 1 $= 6$
no. of 0 $= 4$

Posterior Beta $\Rightarrow (\alpha + w, \beta + n - w)$
(distribution) $(0.5 + 6, 0.5 + 4 - 6)$

$\Rightarrow (6.5, 4.5) \Rightarrow \text{Beta}(6.5, 4.5)$

Posterior Mean $\Rightarrow \frac{\alpha + w}{\alpha + \beta + n}$

$$\alpha' = \alpha + w$$

$$\beta' = \beta + n - w$$

$$\Rightarrow \frac{\alpha'}{\alpha' + \beta'} = \frac{6.5}{6.5 + 4.5}$$

$$\text{or } \frac{0.5 + 6}{0.5 + 0.5 + 10} = \frac{6.5}{11} = \frac{6.5}{11}$$

Normal Distribution Case

Setup: $X_i \sim N(\mu, \sigma^2)$

Prior: $\mu \sim N(\mu_0, \sigma_0^2)$

Posterior Mean: $\hat{\mu} = \frac{n\sigma_0^2}{n\sigma_0^2 + \sigma^2} \bar{X} + \frac{\sigma^2}{n\sigma_0^2 + \sigma^2} \mu_0$

Q. Life (in months) of electric bulbs is normally distributed with unknown mean μ and var. σ^2 . Life of 10 bulbs from a sample are:
2, 2.2, 2.5, 3, 1.8, 2.2, 2.3, 2.7, 2.5, 2.

Find the Bayesian estimator (posterior mean) of μ assuming the Normal(2, 1) Prior distribution.

$$\bar{X} = 2 + 2.2$$

Sol. $\bar{X} = \frac{2 + 2.2 + 2.5 + 3 + 1.8 + 2.2 + 2.3 + 2.7 + 2.5 + 2}{10}$
 $= \frac{23.2}{10} = 2.32$

$\mu_0 = 2$
 $\sigma_0^2 = 1$
 $\sigma^2 = 2$

$$\text{Posterior} = (2.32) \left[\frac{10(1)}{10(1) + 2} \right] + 2 \left[\frac{2}{10(1) + 2} \right] \Rightarrow \frac{23.2}{12} + \frac{4}{12} \Rightarrow \boxed{2.24}$$

Q. Let X be a discrete random var. taking values $\{0, 1, 2\}$ with respective probabilities $\left\{\frac{p}{3}, (1-p), \frac{2p}{3}\right\}$, where $0 \leq p \leq 1$ is a parameter. Consider the samples $2, 1, 1, 0, 0, 2, 1, 2, 0, 0$ from X . Using a Uniform $[0, 1]$ prior.

1) Find the the posterior distribution of p .

Sol.

2, 1, 1, 0, 0, 2, 1, 2, 0, 0

$\rightarrow 0\text{'s} = 4$
 $\rightarrow 1\text{'s} = 3$
 $\rightarrow 2\text{'s} = 3$

$X \in \{0, 1, 2\}$

$$P(X=0) = p/3$$

$$P(X=1) = 1-p$$

$$P(X=2) = \frac{2p}{3}$$

Prior: $p \sim \text{Uniform}(0, 1)$

$\approx p \sim \text{Beta}(1, 1)$

$$\text{Likelihood} \Rightarrow L(p) = \left(\frac{p}{3}\right)^4 (1-p)^3 \left(\frac{2p}{3}\right)^3$$

$$L(p) \propto p^4 (1-p)^3 (p^3) = p^7 (1-p)^3$$

$$\text{Posterior} = L(p) \times \pi(p)$$

$\pi(p|x)$

$$\propto \boxed{p^7 (1-p)^3}$$

since prior is Uniform(0, 1)
 $\pi(p) = 1$

$$f(p) \propto p^{\alpha-1} (1-p)^{\beta-1} \rightarrow \text{Beta Form}$$

$$\alpha = 8, \beta = 4 \rightarrow \text{Beta}(8, 4)$$

Q. Consider i.i.d. bernoulli samples from a Geometric(p) distribution. find the posterior distribution of p using Beta(α, β) prior.

Sol.

$$P(X=x) = (1-p)^{x-1} \cdot p$$

$$\text{Likelihood} = \prod_{i=1}^n (1-p)^{x_i-1} \cdot p = p^n (1-p)^{\sum_{i=1}^n x_i - n}$$

$$\text{Posterior} \propto \text{Prior} \cdot \text{Likelihood}$$

$$\propto [p^{\alpha-1} (1-p)^{\beta-1}] \cdot \left[p^n (1-p)^{\sum_{i=1}^n x_i - n} \right]$$

$$\propto p^{n+\alpha-1} (1-p)^{\sum_{i=1}^n x_i - n + \beta - 1}$$

Ans.

$$\text{Beta} \left(n+\alpha, \beta-n + \sum_{i=1}^n x_i \right)$$

$$\beta = \sum_{i=1}^n x_i - n + \beta$$

$$\alpha = n + \alpha - 1 + 1 = n + \alpha$$